

Title

Effectiveness of a Standalone Digital Platform for Postpartum Depression: A Randomized Comparative Effectiveness Study

Abstract

Postpartum depression affects a substantial proportion of new mothers and limits engagement with usual treatment. We randomized 264 postpartum women to receive either a standalone mobile application designed to address depressive symptoms or an active control intervention. Both groups had access to routine postpartum care. Depressive symptoms were assessed using standardized depression screening tools, and functional capacity was measured using clinician-rated global impression scales at treatment endpoint. The application group demonstrated greater improvement in depression scores compared to controls (mean reduction 8.2 points vs 4.1 points, $p = 0.001$, Cohen $d = 0.55$). Functional impairment also improved more substantially in the intervention group (mean 1.8-point improvement vs 0.9-point improvement on clinical global impression, $p = 0.008$). Treatment effects were observed across diverse demographic subgroups. User engagement with the platform was variable; approximately 58% of participants engaged with core features at least twice weekly. The intervention was well-tolerated with high acceptability ratings. These findings suggest that well-designed standalone digital interventions can meaningfully reduce depressive symptoms in the postpartum period.